BI-STRATEGY

Context:

Capstone Topic: AI-Powered Diabetic Retinopathy Screening and Classification System

The capstone project aims to develop an AI-powered diabetic retinopathy screening system that automatically classifies retinal fundus images into five diabetic retinopathy severity levels. The intended decision-makers are the startup's CEO and healthcare operations manager, who require data-driven insights to improve diagnostic accuracy, screening efficiency, and healthcare service delivery.

KPIs Recommended:

1. Model Accuracy

Percentage of retinal images correctly classified into their respective diabetic retinopathy grades.

2. Sensitivity (Recall)

Measures the system's ability to correctly identify patients affected by diabetic retinopathy.

3. Screening Turnaround Time

Average time required to process a retinal image and generate a diagnosis.

4. False Negative Rate

Percentage of diabetic retinopathy cases incorrectly classified as healthy cases.

5. Screening Coverage

Percentage of registered diabetic patients screened by the system.

6. System Availability

Percentage of time the screening platform remains operational.

7. Cross-Dataset Generalization Score

Measures model performance on unseen datasets and indicates robustness in real-world deployment.

Data Sources:

Retinal fundus images and diagnosis labels are obtained from APTOS 2019, Messidor-2, and partner hospital datasets. Prediction outputs and confusion matrices provide model accuracy, sensitivity, and false negative rates. Application logs capture screening turnaround time and system availability. Patient registration databases help calculate screening coverage. External validation datasets and deployment feedback from hospitals are used to evaluate cross-dataset generalization and real-world model performance.

Tool Stack Recommended:

Data Collection: Hospital information systems, Kaggle datasets, secure uploads

Storage: Google Drive and SQLite

Data Processing: Python, Pandas, NumPy, OpenCV

AI Development: TensorFlow and Keras

Data Analysis: Jupyter Notebook and SQL

Visualization: Looker Studio and Power BI Free

Experiment Tracking: MLflow

Version Control: Git and GitHub

Deployment: Streamlit and Docker

This free-tier technology stack provides an affordable environment for data management, analytics, visualization, and AI model deployment.

Phased Implementation Plan:

Phase 1

Establish data collection pipelines, organize retinal image datasets, and define KPI measurement standards.

Phase 2

Develop and train the diabetic retinopathy classification model, build dashboards, and automate KPI monitoring.

Phase 3

Deploy the system in pilot healthcare environments, integrate real-time analytics, continuously monitor KPIs, and improve model performance using deployment feedback and external validation datasets.

BI Case Study 1 – Swiggy Delivery Analytics

Company:

Swiggy

Business Problem:

Swiggy needed to deliver food quickly while managing thousands of daily orders, fluctuating demand, restaurant preparation times, and delivery partner availability. Delayed deliveries could negatively affect customer satisfaction and retention.

KPIs Tracked:

- **Average Delivery Time (ADT):** Average time taken from order placement to delivery.
- **Order Completion Rate:** Percentage of successfully completed orders.
- **Delivery Partner Utilization:** Number of orders completed per delivery partner.
- **On-Time Delivery Percentage:** Percentage of orders delivered within the estimated time.
- **Customer Satisfaction Rating:** User ratings and feedback scores.
- **Order Cancellation Rate:** Percentage of cancelled orders.

Data Sources Used:

- Customer order transactions
- Restaurant order preparation data
- GPS location data from delivery partners
- Customer feedback and ratings
- Traffic and location information

Tool Stack:

- Python for analytics and machine learning
- Apache Kafka for real-time data streaming
- Data warehouses and cloud storage systems
- SQL for querying large datasets
- Dashboarding and visualization tools for monitoring KPIs

Business Decision/Outcome:

Swiggy uses predictive analytics to forecast demand and optimize delivery partner allocation. Real-time analytics enables efficient routing and dynamic order assignment, reducing delivery times and improving customer experience. Data-driven decisions have helped Swiggy scale operations while maintaining service quality.

Source:

Bytes by Swiggy

BI Case Study 2 – Flipkart Business Intelligence

Company

Flipkart

Business Problem

Flipkart manages millions of products and customers. The company needed to understand customer preferences, improve product recommendations, optimize inventory management, and increase conversions during high-traffic events such as sales campaigns.

KPIs Tracked

- **Conversion Rate:** Percentage of visitors who make purchases.
- **Average Order Value (AOV):** Average spending per order.
- **Cart Abandonment Rate:** Percentage of users who leave without completing purchases.
- **Customer Retention Rate:** Percentage of returning customers.
- **Inventory Turnover Ratio:** Speed at which products are sold and replenished.
- **Recommendation Click-Through Rate (CTR):** Percentage of users interacting with recommended products.

Data Sources Used

- Website clickstream data
- Customer transaction records
- Search and browsing history
- Inventory and supply chain databases
- Customer reviews and ratings
- Marketing campaign data

Tool Stack

- Python for data processing and machine learning
- Apache Spark for large-scale data analytics
- SQL and data warehouses

- Business intelligence dashboards
- Recommendation engines and predictive analytics systems

Business Decision/Outcome

Flipkart uses BI and machine learning to personalize recommendations, forecast demand, and optimize inventory allocation. Analytics-driven recommendations increase customer engagement and conversions, while demand forecasting reduces stock shortages and improves operational efficiency.

Source

Flipkart Tech Blog

BI Maturity Model Mapping

Chosen Sector:

AI-Powered Diabetic Retinopathy (DR) Screening Startup

BI Maturity Matrix:

Dimension	Emerging	Developing	Established	Leading
Data Availability	Basic datasets, mostly unstructured	✔ Retinal images, patient records, prediction logs available; some external data missing	Integrated multi-hospital datasets	Real-time, unified healthcare data ecosystem
Tooling	Excel and manual reports	✔ Python, SQL, dashboards, machine learning models	Automated pipelines and advanced analytics	Real-time AI analytics and MLOps platform
Team	No dedicated analysts	✔ Small team of AI engineers and healthcare professionals	Dedicated BI and data engineering teams	Centre of Excellence with data scientists and domain experts
Use of Insights	Reports rarely used	✔ Insights used for model improvement and screening decisions	Embedded in operational decisions	Analytics drives business strategy and innovation

---

Overall Maturity Stage: Developing

The AI-powered diabetic retinopathy screening startup is currently at the **Developing** stage of BI maturity. The organization collects structured healthcare data, including retinal images and screening results, and uses analytical tools such as Python, SQL, dashboards, and machine learning models to generate insights. A small multidisciplinary team actively uses these insights to improve diagnostic performance and screening workflows. However, enterprise-wide data integration, dedicated BI teams, and strategy-driven analytics capabilities are still evolving, preventing the organization from reaching higher maturity stages.

Quadrant Representation:

Emerging $\longrightarrow$ Developing $\longrightarrow$ Established $\longrightarrow$ Leading

$\uparrow$

AI-Powered Diabetic Retinopathy Screening Startup